9 **(a)** A radiation detector is placed close to a radioactive source.

The variation with time *t* of the measured count rate is shown in Fig. 9.1.

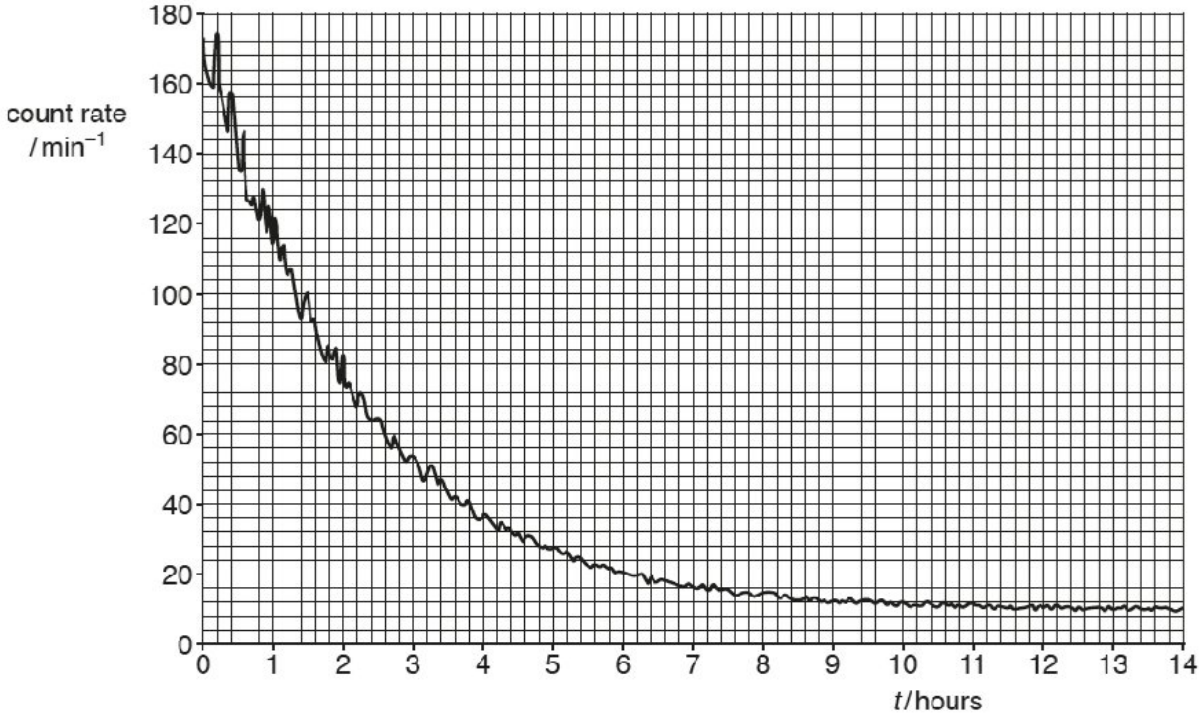


Fig. 9.1

The graph is used to determine the half-life of the radioactive source.

(i) Explain how to take into account the random nature of count rate in the graph in Fig. 9.1.

.....

..... [1]

(ii) Explain how to determine the background count rate from the graph in Fig. 9.1.

.....

..... [1]

(iii) Hence use Fig. 9.1 to determine the half-life of the radioactive source.

half-life = ... hours [3]

(iv) The readings in Fig. 9.1 were obtained at room temperature. A second sample of this radioactive source is heated to a temperature of 500 °C. The initial count rate at time $t = 0$ is the same as that in Fig. 9.1. The variation with time t of the measured count rate from the heated source is determined.

State and explain whether the half-life of the radioactive source changes when the temperature is increased to 500 °C.

.....

.....

..... [2]

(b) An unstable nucleus of nucleon number (mass number) A undergoes α-decay, as illustrated in Fig. 9.2.

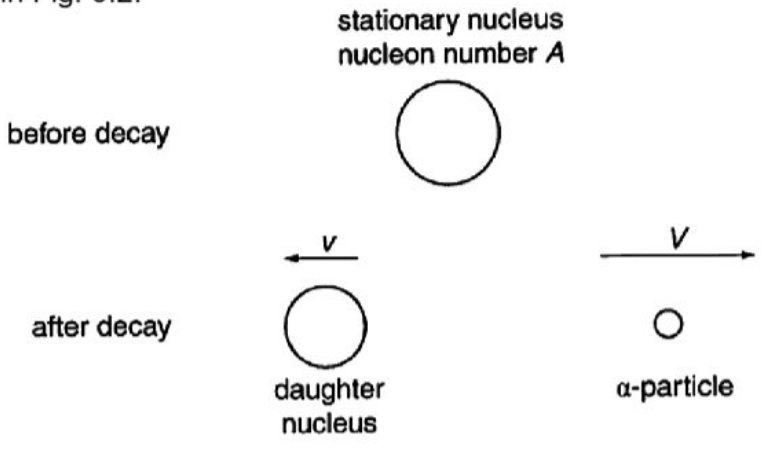


Fig. 9.2

The nucleus is stationary before the decay. After the decay, the initial speed of the α-particle is V and that of the daughter nucleus is v.

(i) State an equation, in terms of A, v and V, to represent conservation of linear momentum for this decay.

.....

..... [1]

(ii) Show that the ratio

$$\frac{\text{initial kinetic energy of } \alpha\text{-particle}}{\text{initial kinetic energy of daughter nucleus}}$$

is equal to $(\frac{1}{4}\ A - 1)$.

[3]

(c) Data for the α-decay of bismuth-212 ($^{212}_{83}Bi$) to form thallium-208 ($^{208}_{81}Tl$) are given in Fig. 9.3.

nucleus	mass of nucleus / u
bismuth-212	211.9459
thallium-208	207.9374
helium-4	4.0015

Fig. 9.3

(i) Use the data of Fig. 9.3 to calculate, to two decimal places, the energy released during the decay.

energy = ... MeV [4]

(ii) Use your answer in **(c)(i)** to show that, based on the expression in **(b)(ii)**, the energy of the α-particle is 6.42 MeV.

[2]

(d) In practice, the α-particle is found to have an energy of 6.10 MeV, rather than the 6.42 MeV shown in **(c)(ii).**

Suggest

(i) an explanation for the difference in energy,

……

..... [1]

(ii) why is it unlikely that the thallium nucleus and the α-particle move off in opposite directions.

……

……

……

..... [2]

[Total: 20]

End of Paper